MANAGEMENT ACCOUNTING

8th EDITION

BY

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PowerPoint Presentation by

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6 PRODUCT & SERVICE COSTING

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

LEARNING OBJECTIVES

1. Describe basic characteristics of & differences between job-order & process costing; identify types of firms that would use each method.
2. Describe cost flows associated with job-order costing.
3. Describe cost flows associated with process costing.
4. Describe equivalent units & explain their role in process costing.

Continued

LEARNING OBJECTIVES

5. Prepare departmental production report using weighted average method.
6. Explain how process costing is affected by nonuniform application of manufacturing inputs & existence of multiple processing departments.
7. Complete departmental production report using FIFO method *(Appendix A)*.
8. Prepare journal entries associated with job-order & process costing *(Appendix B)*.

Click the button to skip Questions to Think About

QUESTIONS TO THINK ABOUT:
Healthblend Nutritional Supplements

Why do you suppose that Brianna did not originally implement an accounting system that would give individual product costs?

Healthblend

QUESTIONS TO THINK ABOUT:
Healthblend Nutritional Supplements

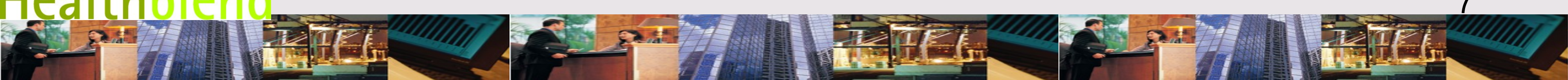
Using a separate work-in-process account for each producing department, describe the flow of costs through Healthblend's plant.

Healthblend

QUESTIONS TO THINK ABOUT:
Healthblend Nutritional Supplements

What types of managerial decisions would be facilitated by having unit product cost information?

Healthblend



QUESTIONS TO THINK ABOUT:
Healthblend Nutritional Supplements

How would Delia's cost accounting system differ from that of Healthblend?

Healthblend

LEARNING OBJECTIVE

1

Describe basic characteristics of & differences between job-order & process costing; identify types of firms that would use each method.

JOB-ORDER COSTING:
Definition

An accounting system that assigns costs to products produced for individually specific jobs.

JOB-ORDER COSTING

The key feature of job-order costing is that the cost of 1 job differs from that of another and must be tracked separately.

LO 1

PROCESS COSTING: Definition

An accounting system that assigns costs to products produced in a series of processes.

JOB-ORDER COSTING

The key feature of process costing is that the products produced are homogeneous and therefore have the same cost.

JOB-ORDER: Product Costs

- Direct materials
- Direct labor
- Overhead applied at predetermined rate

LEARNING OBJECTIVE

2

Describe cost flows associated with job-order costing.

How do you calculate costs for a job-order cost system?

Combine direct materials + direct labor + overhead.

SUPLISHAKE-001

Materials cost $1,780

Direct labor $300 (20 hours x $15)

Overhead $240 (20 hours x $12)

Job 001

Materials	**$1,780**
Labor	**300**
Overhead	**240**
Total	**$2,320**
Unit cost ($2,320/200)	**$11.60**

What will be the selling price for SupliShake-001?

If the selling price is cost + 50%, PNP will sell SupliShake-001 for **$3,480** ($2,320 + $1160).

WORK-IN-PROCESS: Definition

All incomplete work at the end of an accounting period.

LO 2

ACCOUNTING FOR OVERHEAD

Annual overhead costs (depreciation, rent, utilities, insurance) are estimated to be $14,400

Overhead rate

= $14,400 / 1,200 direct labor hours

= $12 per DLH

LIGASTRONG-001

Materials cost $1,300

Direct labor $450 (30 hours x $15)

Overhead $360 (30 hours x $12)

Job 001

Materials	**$1,780**
Labor	**450**
Overhead	**360**
WIP 50% complete . .	**$2,590**

OVERHEAD

Overhead is underapplied by $15 for month.

Actual Overhead Costs	
Rent	$ 400
Utilities	50
Depreciation.	100
Insurance	65
Total Overhead Cost .	$ 615

COST OF GOODS SOLD

Underapplied overhead adjusts CGS.

Statement of Cost of Goods Sold

Beginning finished goods inventory	$ 0
Cost of goods manufactured	2,320
Goods available for sale	$2,320
Less: Ending finished goods inventory	0
Normal cost of goods sold	$2,320
Add: Underapplied overhead	15
Adjusted cost of goods sold	$2,335

Exhibit 6-10 Statement of Cost of Goods Sold

EXHIBIT 6-10

LEARNING OBJECTIVE

3

Describe cost flows associated with process costing.

HEALTHBLEND: 3 Processes

- Picking Department
  - DL selects herbs, vitamins, minerals, inert materials
  - Ingredients are combined
- Encapsulating Department
  - Mixture loaded into gelatin capsule
- Bottling Department
  - Capsules counted into bottles & labeled

OPERATION COSTING:
Blending Systems

- Operation costing blends job-order and process costing
  - Material costs accumulated by batch
    - Job-order costing applies
  - Labor, overhead costs accumulated by process

LO 3

TYPES OF PROCESS MANUFACTURING

- Sequential processing
  - Materials pass through different process <u>sequentially</u>
  - Example:
    - Bottling follows picking at Healthblend
- Parallel processing
  - Different materials pass through different processes simultaneously

PRODUCTION REPORT:
Definition

Provides information about physical units processed in a department as well as manufacturing costs.

LEARNING OBJECTIVE

4

Describe equivalent units & explain their role in process costing.

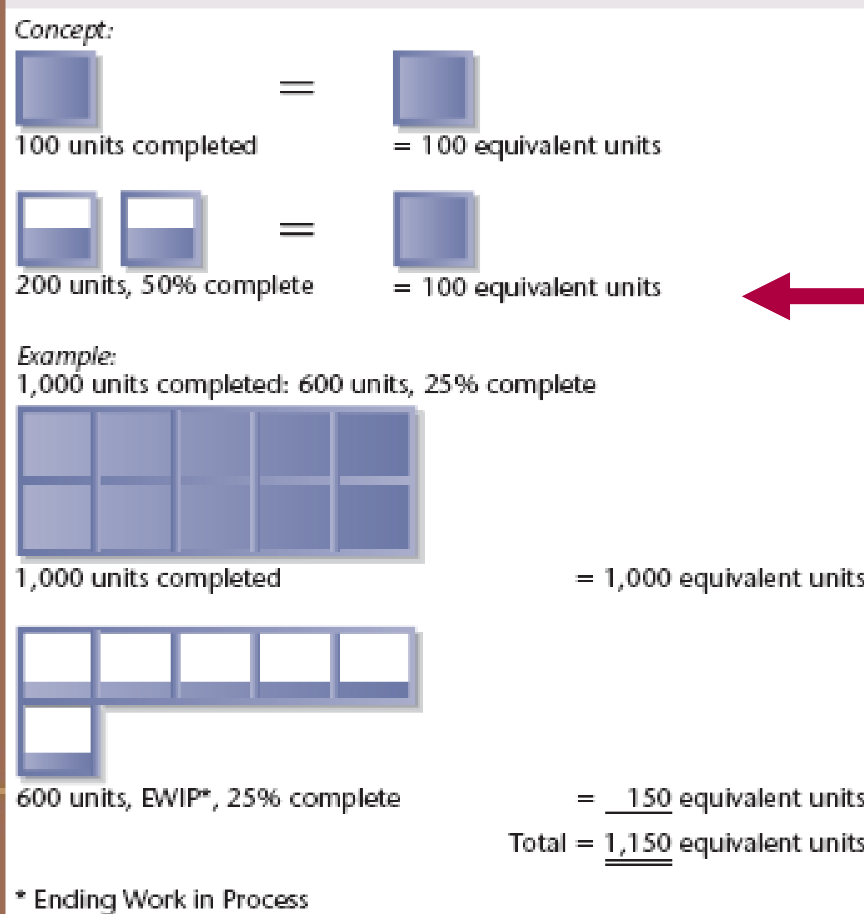
QUANTIFYING WIP IN A PROCESS COST SYSTEM

- Problems counting WIP
  - How to define a unit of production?
    - *Answer*: Equivalent full units (EFU)
  - How should beginning WIP be treated?
    - Weighted average
    - FIFO

EQUIVALENT FULL UNITS:
Definition

The complete units that *could have been produced* given the total amount of manufacturing effort expended for the period.

DEFINING UNIT OF PRODUCTION: Concept



Equivalent full units (EFU) necessary to calculate unit cost.

EXHIBIT 6-13

BEGINNING WORK-IN-PROCESS

- 2 ways to cost beginning WIP
  - Weighted average
    - Combines beginning inventory costs with current period costs
    - Costs are pooled into 1 average unit cost
  - FIFO
    - Separates beginning inventory costs from current period costs

LEARNING OBJECTIVE

5

Prepare departmental production report using *weighted average method*.

5 STEPS TO PREPARE PRODUCTION REPORT

1. Physical units flow analysis
2. Calculation of equivalent units
3. Computation of unit cost
4. Valuation of inventories
   a. Goods transferred out
   b. Ending work in process
5. Cost reconciliation

HEALTHBLEND'S PICKING DEPT.: July Costs

Production	
Units in process July 1, 75% complete	20,000
Units complete & transferred out	50,000
Units in process July 31, 25% complete	10,000
Costs	
Work in process, July 1	$ 3,525
Cost added during July	10,125

CALCULATE EFU: Step 2

In Step 1, calculate units to be accounted for. Then calculate EFU.

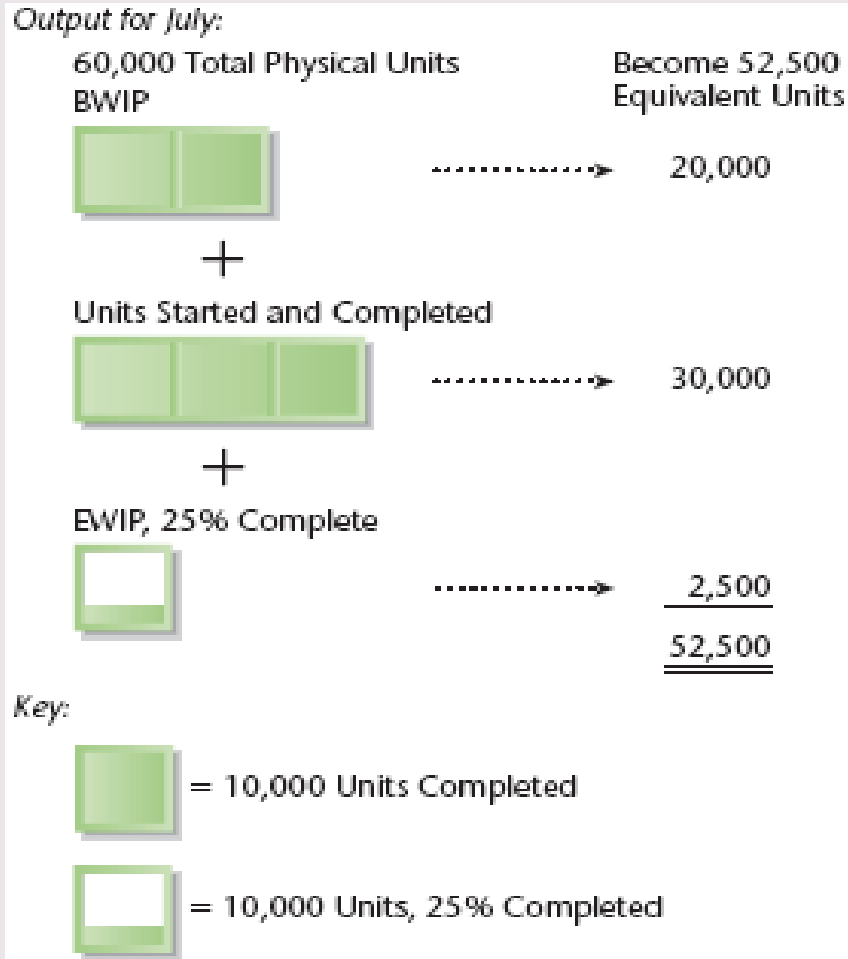


EXHIBIT 6-14

COMPUTE UNIT COST: Steps 3 & 4

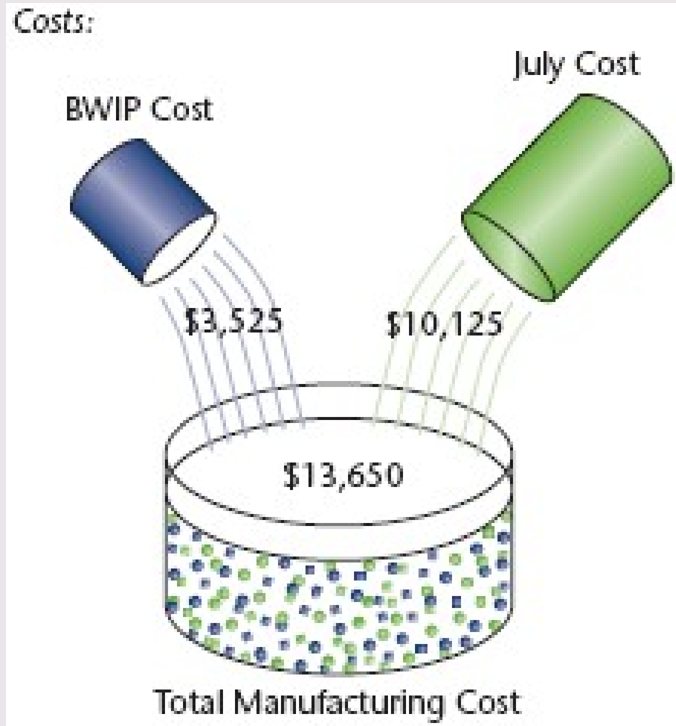
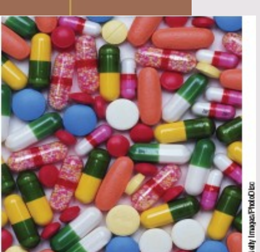


EXHIBIT 6-14



Unit cost = $13,625 / 52,500 = $0.26 per EFU

Transferred out ($0.26 x 50,000 = $13,000

EWIP ($0.26 x 2,500) = 650

Total cost assigned = $13,650

COST RECONCILIATION: Step 5

Total Manufacturing Costs Assigned	
Goods transferred out	$ 13,000
Goods in ending WIP	650
Total costs accounted for	$ 13,650
Manufacturing Costs to Account For	
Beginning WIP	$ 3,525
Incurred during the period	10,125
Total costs to account for	$ 13,650

WEIGHTED AVERAGE:

Evaluation

- Major benefit
  - Simplicity
- Major disadvantage
  - Accuracy in computing unit costs for current period & for beginning WIP

LEARNING OBJECTIVE

6

Explain how process costing is affected by nonuniform application of manufacturing inputs & existence of multiple processing departments.

NONUNIFORM INPUTS:
Definition

Occurs, for example, when direct materials are completely added at the *beginning of process* rather than *throughout process*.

LO 6

NONUNIFORM INPUTS: Example

In the Bottling Department, filled capsules, bottles added at *beginning* but bottle caps, boxes added at *end* of process. Affects computation of EFU.

LEARNING OBJECTIVE

7

Complete departmental production report using FIFO method *(Appendix A)*.

HEALTHBLEND'S PICKING DEPT.: July Costs

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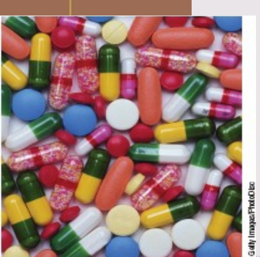

Remember

COST RECONCILIATION: Step 5

Goods transferred out	
Units in beginning WIP	$ 4,875
Units started & completed	8,100
Goods in ending WIP	675
Total costs accounted for	$ 13,650
Manufacturing Costs to Account For	
Beginning WIP	$ 3,525
Incurred during the period	10,125
Total costs to account for	$ 13,650

LEARNING OBJECTIVE

8

Prepare journal entries associated with job-order & process costing *(Appendix B)*.

How are transactions entered into the accounting system?

Transactions are entered into accounting system by making journal entries & posting to accounts.

CHAPTER 6

THE END